Tender Ref ID: 999239 | Notice No: 23/MD/CE-Assembly/AGICL/2026 , Dated: 28.09.2026
Estimated Contract Value: Rs. 3,62,94,39,875/- | Bid Closing: 29/10/2026 03:00 PM
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STRUCTURAL DESIGN BASIS REPORT · DESIGN CRITERIA & CODES

1. STRUCTURAL DESIGN CODES & STANDARDS:
? IS 456:2000 ? Code of Practice for Plain and Reinforced Concrete
? IS 800:2007 ? General Construction in Steel ? Code of Practice
? IS 875 (Part 3): 2015 ? Wind Loads on Buildings and Structures (Basic Wind Speed: 50 m/s for Amaravati region)
? IS 1893 (Part 1): 2016 ? Criteria for Earthquake Resistant Design (Seismic Zone III, Importance Factor I = 1.5)
? IS 13920:2016 ? Ductile Design and Detailing of Reinforced Concrete Structures

2. MATERIAL STRENGTH PARAMETERS:
? Concrete Characteristic Compressive Strength (fck): M50 (50 MPa) for raft; M60 (60 MPa) for columns and shear walls.
? Structural Steel Yield Strength (fy): 355 MPa (S355) and 460 MPa (S460).
? Rebar Yield Strength (fy): 550 MPa (Fe550D).

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STRUCTURAL DESIGN BASIS REPORT · ETABS 3D FINITE ELEMENT MODEL

3. 3D FINITE ELEMENT MODELING & ANALYSIS (ETABS / SAP2000):
? Geometry: Complete 3D spatial simulation incorporating non-linear P-Delta second-order geometric effects.
? Construction Staging: Full construction sequence simulation modeled across 24 erection stages.
? Wind Tunnel Calibration: High-frequency force balance (HFFB) test data from RWDI integrated into response spectra.

4. STRUCTURAL PERFORMANCE BENCHMARKS:
? Maximum Lateral Top Deflection under 50-year return wind: 480mm (H/520, well within allowable H/500).
? Maximum Story Drift: 0.002h under unfactored wind and seismic loading.
? Dynamic Natural Period: T1 = 3.57 sec (lateral sway), T2 = 3.42 sec (orthogonal sway), T3 = 1.85 sec (torsion).

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STRUCTURAL DESIGN BASIS REPORT · FOUNDATION & GEOTECHNICAL

5. GEOTECHNICAL & FOUNDATION CRITERIA:
? Subsurface Strata: Silty sand overlying hard fractured granite bedrock encountered at depth of 18m to 24m.
? Foundation Type: Pile-Raft composite system with 1200mm diameter bored cast-in-situ concrete piles.
? Pile Socketing: Minimum 3.0-meter rock socket into sound granite (RQD > 60%).
? Safe Working Capacity per Pile: 850 MT in compression; 180 MT in tension (uplift resistance).